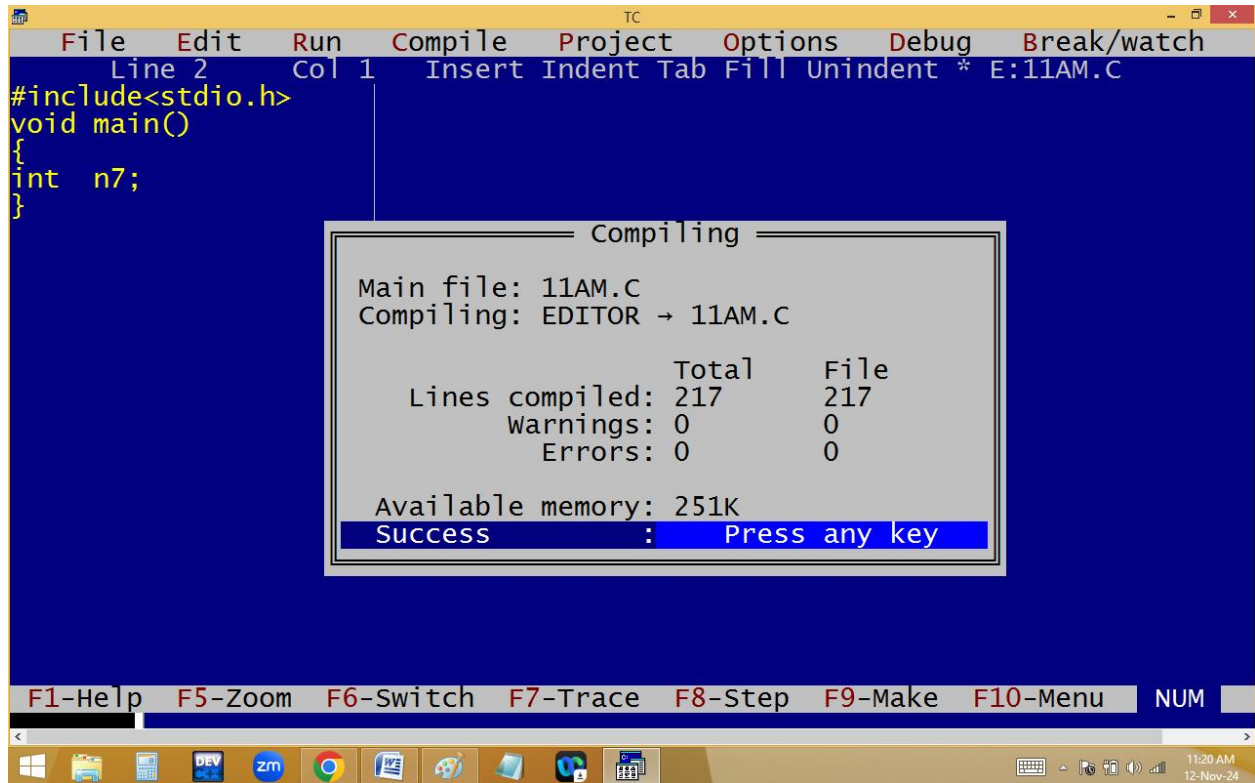


Identifier naming rules:

1. Name should have to start with alphabet or underscore [_] only.



The screenshot shows the Turbo C++ IDE with a C program being compiled. The program code is as follows:

```
Line 2 Col 1
#include<stdio.h>
void main()
{
int n7;
}
```

The compilation window displays the following information:

Compiling

Main file: 11AM.C
Compiling: EDITOR → 11AM.C

	Total	File
Lines compiled:	217	217
Warnings:	0	0
Errors:	0	0

Available memory: 251K
Success : Press any key

The IDE interface includes a menu bar (File, Edit, Run, Compile, Project, Options, Debug, Break/watch), a toolbar, and a status bar with function key shortcuts (F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, F10-Menu, NUM). The Windows taskbar at the bottom shows the time as 11:20 AM on 12-Nov-24.

TC

File Edit Run Compile Project Options Debug Break/watch

Line 4 Col 8 Insert Indent Tab Fill Unindent * E:11AM.C

```
#include<stdio.h>
void main()
{
int 7n;
}
```

Compiling

Main file: 11AM.C
Compiling: EDITOR → 11AM.C

	Total	File
Lines compiled:	216	216
Warnings:	0	0
Errors:	1	1

Available memory: 251K

Errors : Press any key

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Menu NUM

11:21 AM 12-Nov-24

TC

File Edit Run Compile Project Options Debug Break/watch

Line 4 Col 7 Insert Indent Tab Fill Unindent * E:11AM.C

```
#include<stdio.h>
void main()
{
int _n;
}
```

Compiling

Main file: 11AM.C
Compiling: EDITOR → 11AM.C

	Total	File
Lines compiled:	217	217
Warnings:	0	0
Errors:	0	0

Available memory: 251K

Success : Press any key

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Menu NUM

11:21 AM 12-Nov-24

2. Numbers allowed but not at first position.

3. Spaces not allowed.

```
File Edit Run Compile Project Options Debug Break/watch
Line 4 Col 9 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
void main()
{
int abc;
}
```

Compiling

Main file: 11AM.C
Compiling: EDITOR → 11AM.C

	Total	File
Lines compiled:	217	217
Warnings:	0	0
Errors:	0	0

Available memory: 251K
Success : Press any key

Message

•Compiling E:\TC\11AM.C:

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Menu NUM

```
File Edit Run Compile Project Options Debug Break/watch
Line 4 Col 11 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
void main()
{
int a b c;
}
```

Compiling

Main file: 11AM.C
Compiling: EDITOR → 11AM.C

	Total	File
Lines compiled:	216	216
Warnings:	0	0
Errors:	1	1

Available memory: 251K
Errors : Press any key

Message

•Compiling E:\TC\11AM.C:

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Menu NUM

4. No special character except underscore.

Line 4 Col 10 Insert Indent Tab Fill Unindent * E:11AM.C

```
#include<stdio.h>
void main()
{
int a$b ;
}
```

Compiling

Main file: 11AM.C
Compiling: EDITOR → 11AM.C

	Total	File
Lines compiled:	216	216
Warnings:	0	0
Errors:	2	2

Available memory: 250K
Errors : Press any key

Message

•Compiling E:\TC\11AM.C:
Error E:\TC\11AM.C 4: Declaration syntax error in function main

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Menu NUM

Line 4 Col 15 Insert Indent Tab Fill Unindent * E:11AM.C

```
#include<stdio.h>
void main()
{
int _a_b_ ;
}
```

Compiling

Main file: 11AM.C
Compiling: EDITOR → 11AM.C

	Total	File
Lines compiled:	217	217
Warnings:	0	0
Errors:	0	0

Available memory: 250K
Success : Press any key

Message

•Compiling E:\TC\11AM.C:

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Menu NUM

5. Keywords not allowed as identifiers.

TC

File Edit Run Compile Project Options Debug Break/watch

Line 4 Col 11 Insert Indent Tab Fill Unindent * E:11AM.C

```
#include<stdio.h>
void main()
{
int while ;
}
```

Compiling

Main file: 11AM.C
Compiling: EDITOR → 11AM.C

	Total	File
Lines compiled:	216	216
Warnings:	0	0
Errors:	1	1

Available memory: 250K
Errors : Press any key

Message

•Compiling E:\TC\11AM.C:

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Menu NUM

11:24 AM 12-Nov-24

TC

File Edit Run Compile Project Options Debug Break/watch

Line 4 Col 9 Insert Indent Tab Fill Unindent * E:11AM.C

```
#include<stdio.h>
void main()
{
int for ;
}
```

Compiling

Main file: 11AM.C
Compiling: EDITOR → 11AM.C

	Total	File
Lines compiled:	216	216
Warnings:	0	0
Errors:	1	1

Available memory: 250K
Errors : Press any key

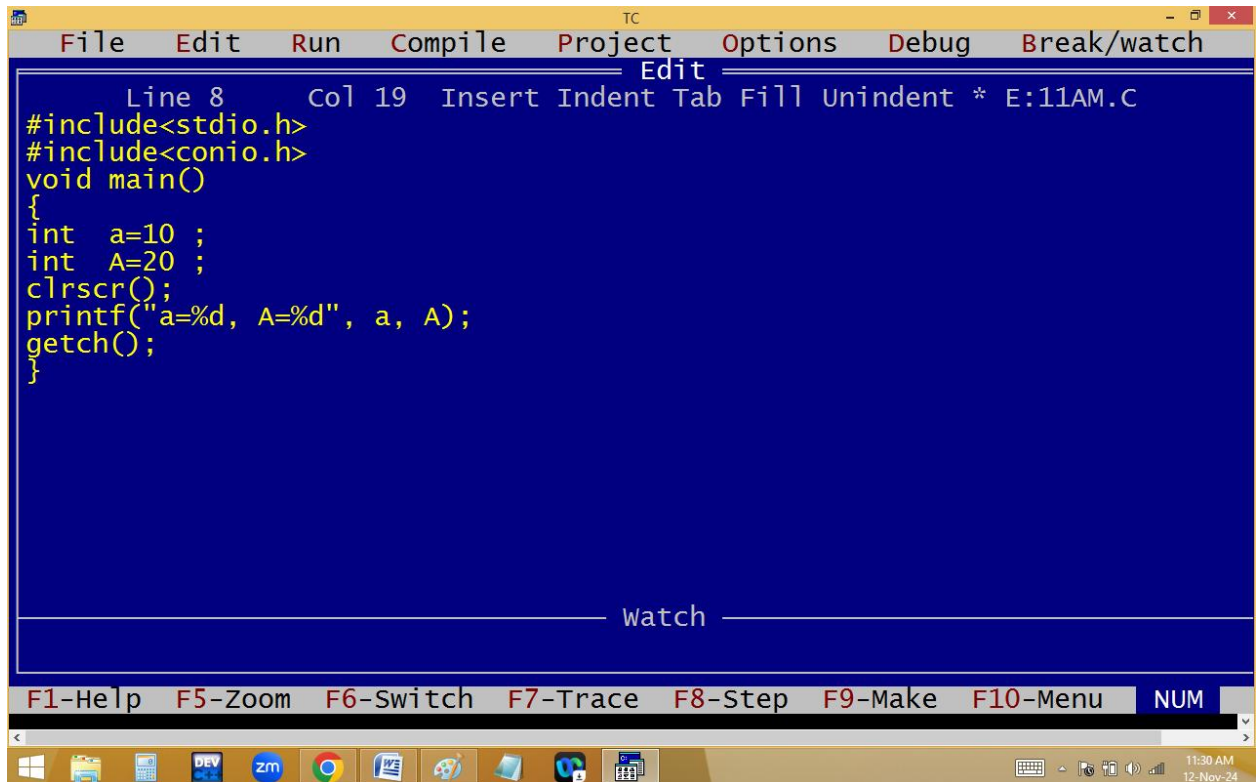
Message

Compiling E:\TC\11AM.C:
•Error E:\TC\11AM.C 4: Declaration syntax error in function main

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Menu NUM

11:25 AM 12-Nov-24

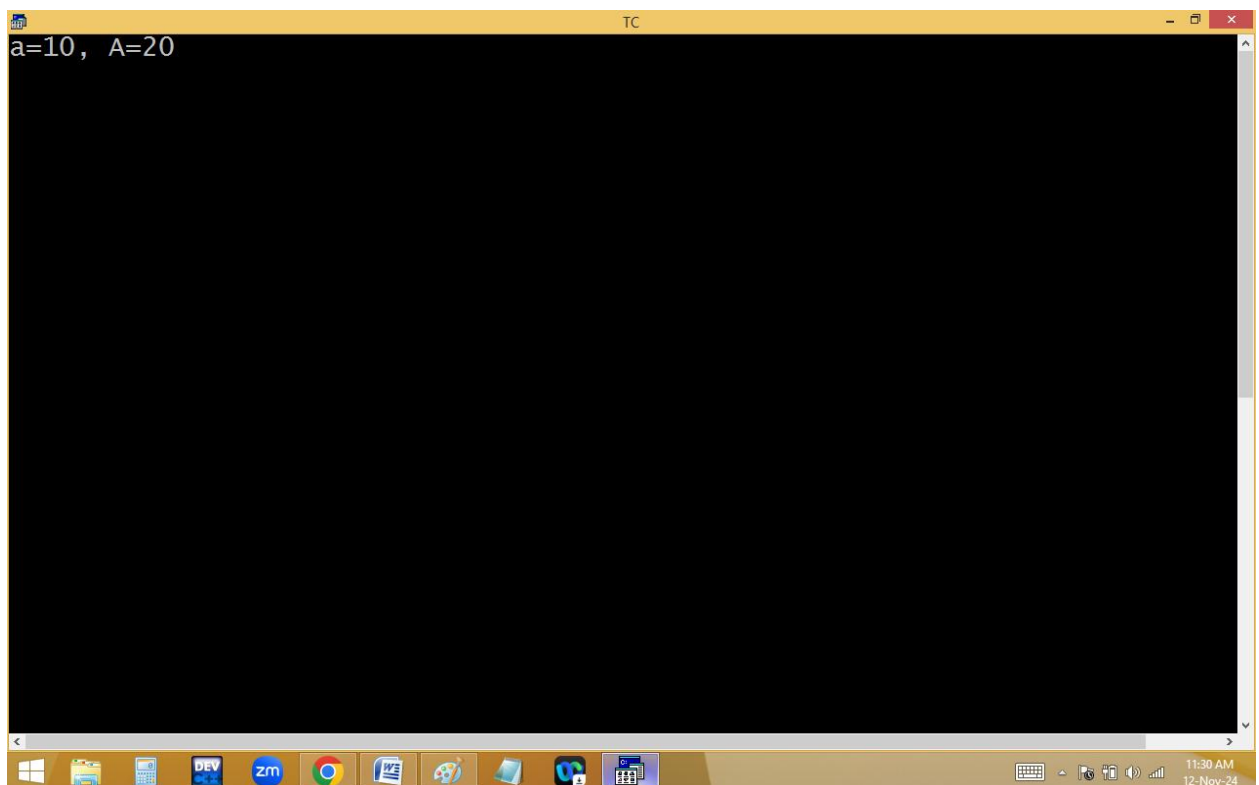
6. Identifiers are case sensitive i.e. lower and upper are different.



The screenshot shows the Turbo C++ IDE with a yellow title bar and a menu bar. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The 'Edit' menu is open, showing options like Line 8, Col 19, Insert, Indent, Tab, Fill, Unindent, and a timestamp * E:11AM.C. The main editor area has a blue background and contains the following C code:

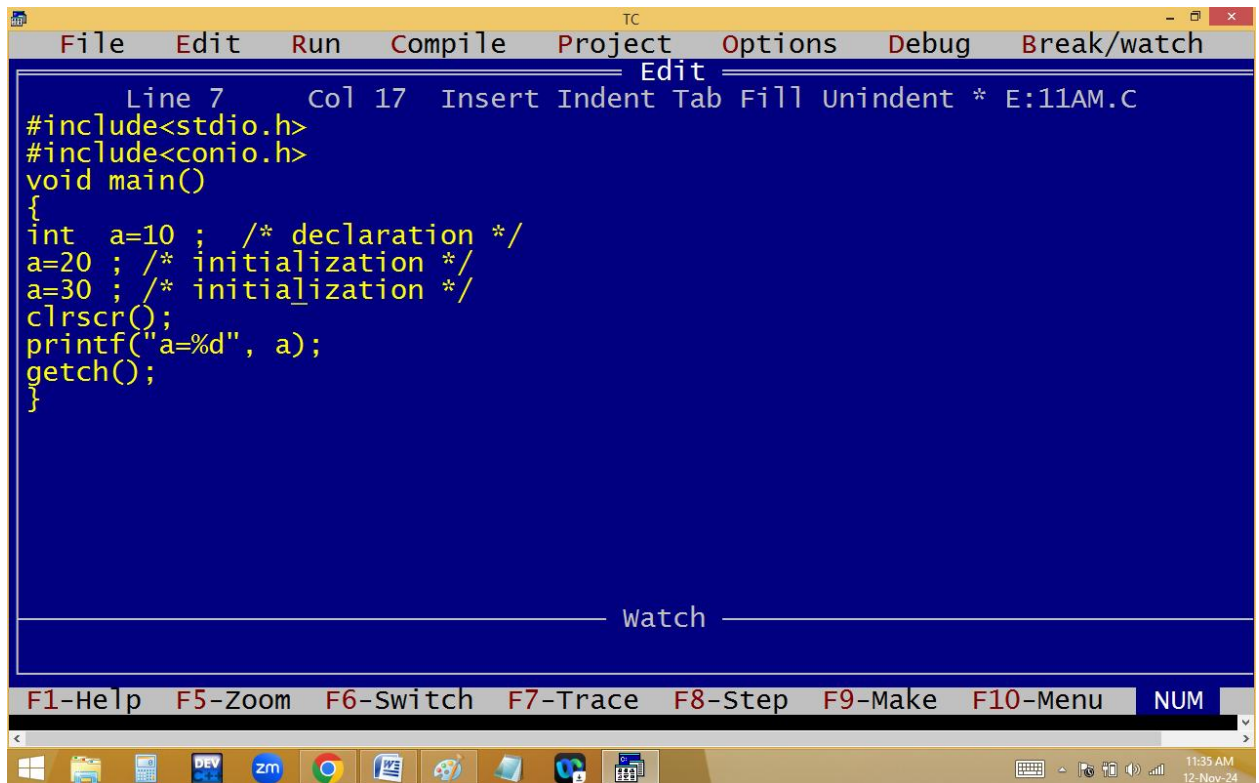
```
#include<stdio.h>
#include<conio.h>
void main()
{
int  a=10 ;
int  A=20 ;
clrscr();
printf("a=%d, A=%d", a, A);
getch();
}
```

Below the editor is a 'Watch' section. At the bottom of the IDE is a toolbar with icons for F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, F10-Menu, and NUM. The Windows taskbar is visible at the very bottom with various application icons and a system clock showing 11:30 AM on 12-Nov-24.



The screenshot shows the Turbo C++ IDE with a yellow title bar. The main editor area is black and displays the output of the program: "a=10, A=20". The Windows taskbar is visible at the bottom with various application icons and a system clock showing 11:30 AM on 12-Nov-24.

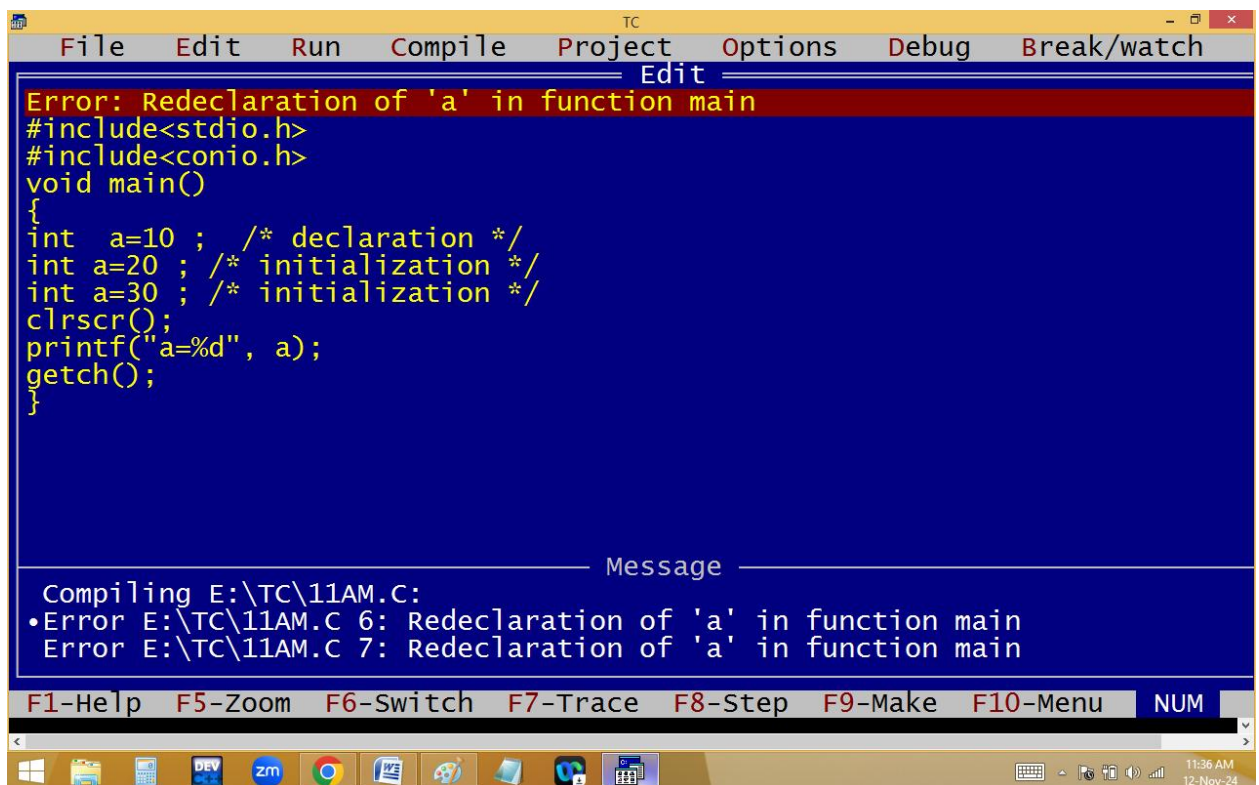
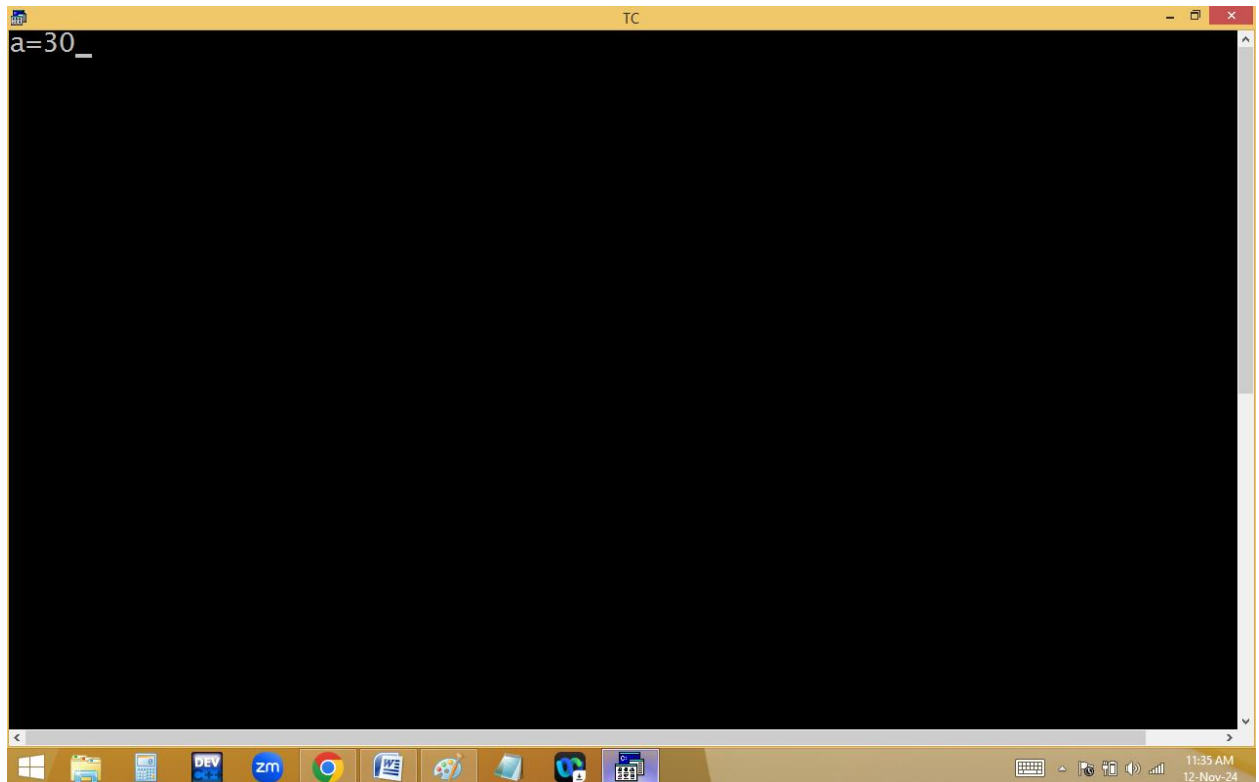
7. Duplicate identifiers are not allowed in same function or block { }



The screenshot shows the Turbo C++ (TC) IDE interface. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 7 Col 17 Insert Indent Tab Fill Unindent * E:11AM.C'. The main editor area contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int a=10 ; /* declaration */
    a=20 ; /* initialization */
    a=30 ; /* initialization */
    clrscr();
    printf("a=%d", a);
    getch();
}
```

Below the code editor is a 'Watch' section. At the bottom, a toolbar contains function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, F10-Menu, and NUM. The Windows taskbar at the very bottom shows various application icons and the system clock indicating 11:35 AM on 12-Nov-24.



The screenshot shows the Turbo C++ (TC) IDE interface. The title bar reads 'TC'. The menu bar includes 'File', 'Edit', 'Run', 'Compile', 'Project', 'Options', 'Debug', and 'Break/watch'. The 'Edit' menu is currently open. The main text area has a blue background and contains the following C code:

```
Error: Redclaration of 'a' in function main
#include<stdio.h>
#include<conio.h>
void main()
{
    int    a=10 ; /* declaration */
    float  a=2.50 ; /* declaration */
    clrscr();
    printf("a=%d", a);
    getch();
}
```

Below the code area is a 'Message' window. It displays the compilation status: 'Compiling E:\TC\11AM.C:' followed by the error message: '•Error E:\TC\11AM.C 6: Redclaration of 'a' in function main'. At the bottom of the IDE, there is a toolbar with function key shortcuts: 'F1-Help', 'F5-Zoom', 'F6-Switch', 'F7-Trace', 'F8-Step', 'F9-Make', 'F10-Menu', and 'NUM'. The Windows taskbar is visible at the very bottom, showing the Start button, several application icons (including DEV, zm, and Chrome), and the system clock indicating '11:45 AM 12-Nov-24'.

```
TC
File Edit Run Compile Project Options Debug Break/watch
Edit
Line 15 Col 1 Insert Indent Tab Fill Unindent * E:11AM.C
#include<stdio.h>
#include<conio.h>
void house2()
{
int a=1200;
}
void main()
{
int a=10 ; /* declaration */
clrscr();
printf("a=%d", a);
getch();
}

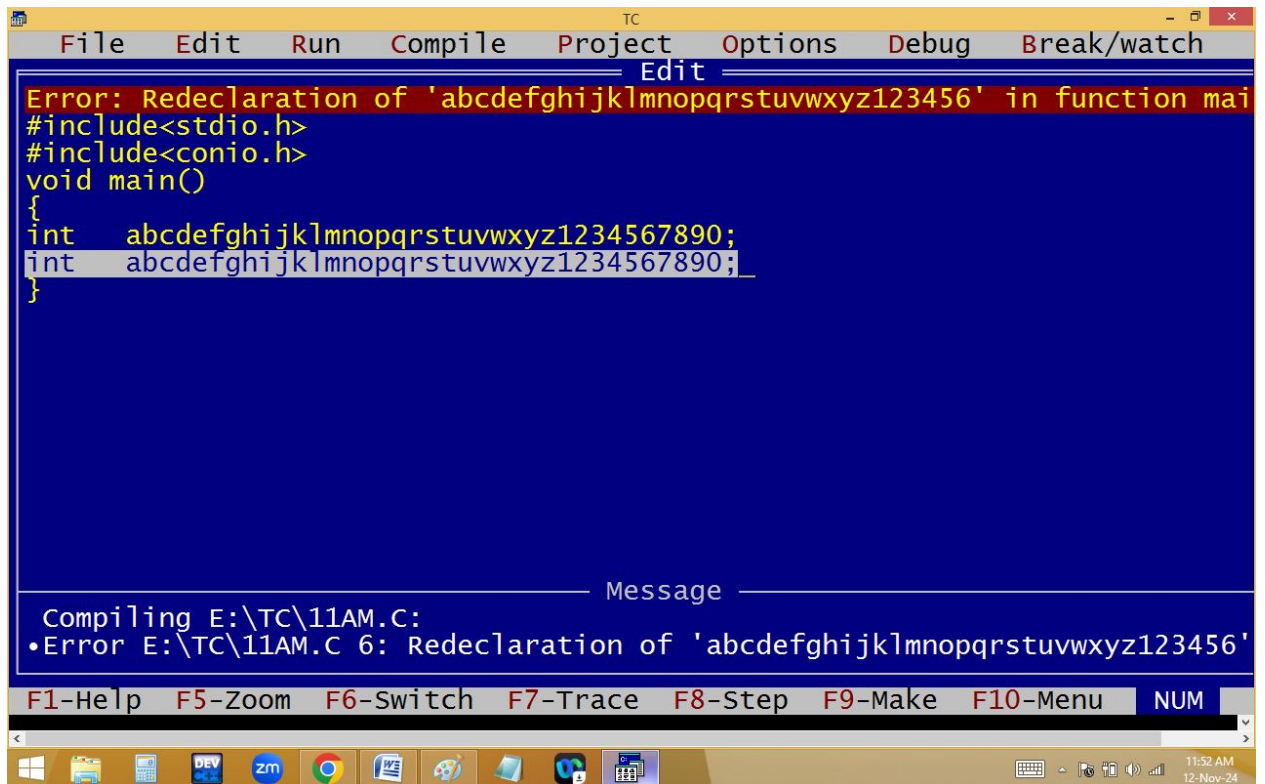
_

Watch

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Menu NUM
```

```
TC
a=10_
```

8. Name may contain up to 32 characters and excess characters ignored by the compiler.



The screenshot shows the Turbo C++ (TC) IDE interface. The menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The main editor window displays the following code:

```
Error: Redeclaration of 'abcdefghijklmnopqrstuvwxyz123456' in function mai
#include<stdio.h>
#include<conio.h>
void main()
{
int  abcdefghijklmnopqrstuvwxyz1234567890;
int  abcdefghijklmnopqrstuvwxyz1234567890;
}
```

The error message at the top indicates a "Redeclaration of 'abcdefghijklmnopqrstuvwxyz123456' in function mai". The variable names in the code are 33 characters long, which exceeds the 32-character limit. The status bar at the bottom shows function key shortcuts: F1-Help, F5-Zoom, F6-Switch, F7-Trace, F8-Step, F9-Make, F10-Menu, and NUM. The Windows taskbar at the very bottom shows the time as 11:52 AM on 12-Nov-24.

Message

Compiling E:\TC\11AM.C:
•Error E:\TC\11AM.C 6: Redeclaration of 'abcdefghijklmnopqrstuvwxyz123456'

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Menu NUM

TC

File Edit Run Compile Project Options Debug Break/watch

Edit

Error: Redclaration of 'abcdefghijklmnopqrstuvwxyz123456' in function main

```
#include<stdio.h>
#include<conio.h>
void main()
{
int   abcdefghijklmnopqrstuvwxyz1234567890;
int   abcdefghijklmnopqrstuvwxyz123456;_
}
```

Message

Compiling E:\TC\11AM.C:
•Error E:\TC\11AM.C 6: Redclaration of 'abcdefghijklmnopqrstuvwxyz123456'

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Menu NUM

11:54 AM 12-Nov-24

TC

File Edit Run Compile Project Options Debug Break/watch

Line 5 Col 1 Insert Indent Tab Fill Unindent * E:11AM.C

```
#include<stdio.h>
void main()
{
int   abcdefghijklmnopqrstuvwxyz1234567890;
int   abcdefghijklmnopqrstuvwxyz12345;
}
```

Compiling

Main file: 11AM.C
Compiling: EDITOR → 11AM.C

	Total	File
Lines compiled:	220	220
Warnings:	0	0
Errors:	0	0

Available memory: 250K
Success : Press any key

F1-Help F5-Zoom F6-Switch F7-Trace F8-Step F9-Make F10-Menu NUM

11:55 AM 12-Nov-24

DATA TYPES

Data type determines the type of value we are going to store in our computer. To store anything in our computer, we should have to allocate the memory. This memory allocation is depended on the data type we are using.

Data type determines the properties such as

1. No of bytes
2. Range
3. Type of value

In C language we are having 3 **basic** data types

- 1. Int – To store non-decimal numbers**
- 2. Float – To store decimal numbers**
- 3. Char – To stores alphabets, numbers and special char**

Total data types are divided into 3 types.

- 1. Primitive data types**
- 2. Derived data types**
- 3. User defined data types**

PRIMITIVE DATA TYPES:

These are the regular data types we are using in our c programs.

Data type	Bytes	Conversion Character / format specifier	Storage Range
int / signed int / short int	2	%d	-32768 to +32767
unsigned int	2	%u	0 to 65535
long int	4	%ld	-2147483648 to 2147483647
unsigned long int	4	%lu	0 to 4294967295
float	4	%f	$3.4 * 10^{-38}$ to $3.4 * 10^{+38}$
double	8	%lf	$1.7 * 10^{-308}$ to $1.7 * 10^{+308}$
long double	10	%Lf	$3.4 * 10^{-4932}$ to $1.1 * 10^{+4932}$
char	1	%c	1 character Signed char [-128 to +127] Unsigned char [0 to 255]
char[10] (STRING)	10	%s	9 char + 1 null char
void [empty data type]			nothing

DERIVED DATA TYPES:

They are derived from primitive data types.

1. Array [non-primitive]
2. Pointer

3. Function

USER DEFINED DATA TYPES:

These are the data types created by the user.

1. structure
2. union
3. enum